



Scope (Local,
Global, Block
Scope)
and
Input/Output in
Python

➤ What is Variable Scope?

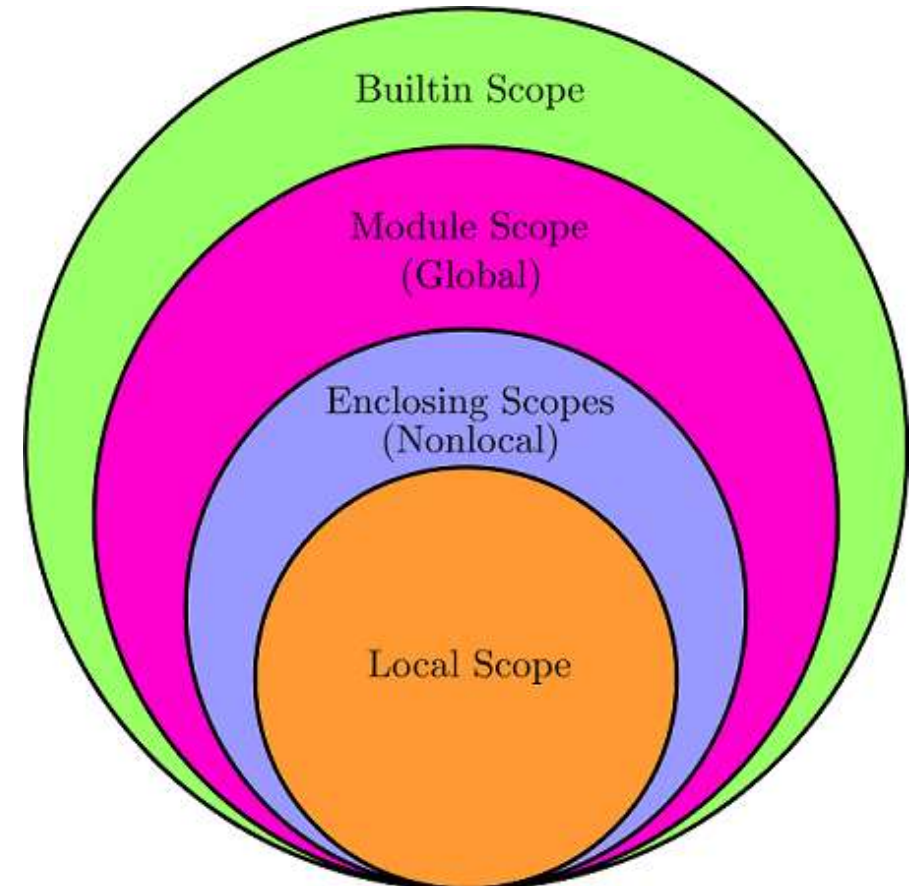
Scope refers to the **visibility and lifetime** of a variable within a program.

🔍 Why it matters?

- Prevents variable conflicts
- Helps in memory management
- Organizes variable access

☐ Python Follows:

LEGB Rule – Local, Enclosing, Global, Built-in



➤ Local Scope (Function Scope)

✦ A variable declared **inside a function** is **local** to that function.

⚠ Access outside the function = Error

□ Example:

```
def greet():
    message = "Hello!"
    print(message)
```

```
greet()          # ✅ Output: Hello!
```

```
print(message)  # ❌ Error: message is not defined
```



🔑 **Note:** Local variables are created when the function is called and destroyed when the function ends.



➤ Global Scope

✦ A variable declared **outside** of any function is **global**.

✓ Accessible inside functions (read-only)

✗ Can't modify unless declared with global keyword

□ Example:

```
name = "Ritesh"
```

```
def show():  
    print(name)
```

```
show() #  Output: Ritesh
```



➤ Modifying Global Variable inside Function

🔑 Use **global keyword** to modify global variables within functions.

□ Example:

```
count = 0

def increment():
    global count
    count += 1

increment()
print(count) # ✅ Output: 1
```

⚠ **Caution:** Overuse of global can make code harder to understand and debug.



➤ Enclosing Scope (Nested Functions)

✦ When a function is **nested inside another**, the inner function can access the **enclosing function's** variables.

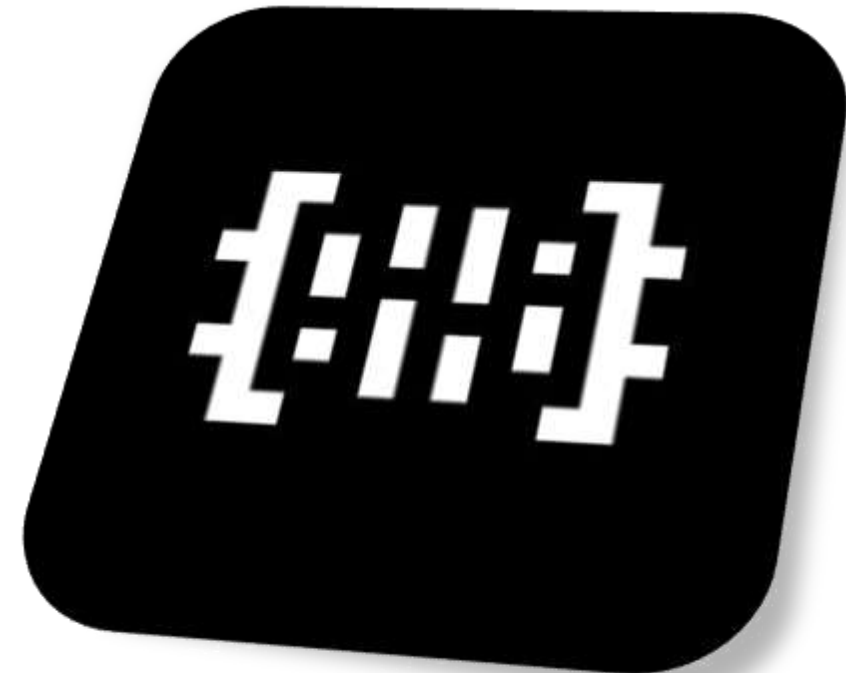
□ **Example:**

```
def outer():
    msg = "Enclosed"

    def inner():
        print(msg)

    inner()

outer() #  Output: Enclosed
```



🔍 **Also known as:** nonlocal scope



LEGB Rule – Scope Resolution Order

Python resolves variables using **LEGB**:

Level	Description
Local	Inside the current function
Enclosing	Inside enclosing (outer) function
Global	At the top level of the module
Built-in	Predefined names in Python



➤ Block Scope in Python?

✗ Python does not have block-level scope

In C/C++/Java, variables declared in {} are block-scoped.

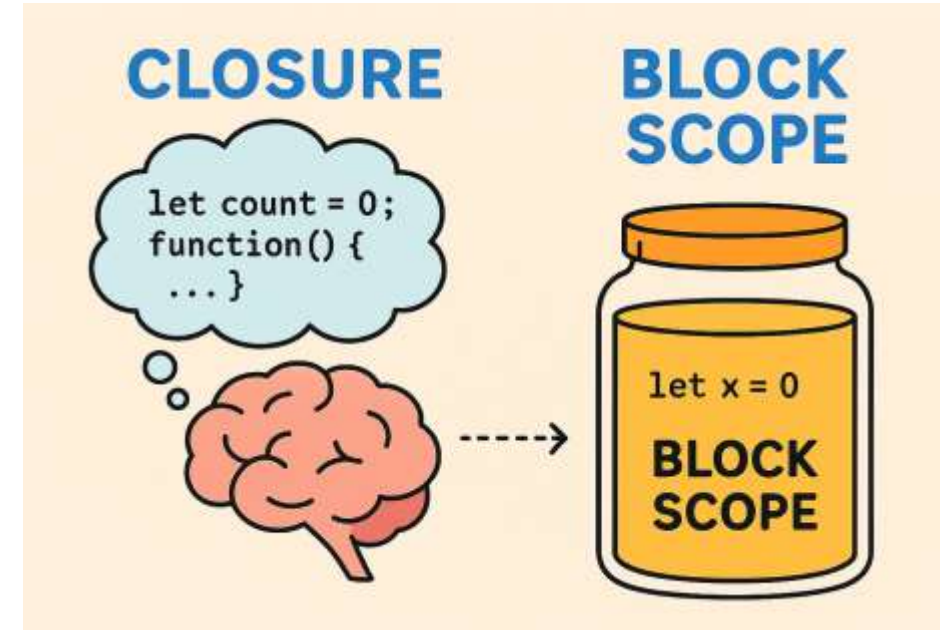
□ Example in Python:

```
if True:
    x = 100
print(x)
```

✓ Output: 100

🔍 Why?

Because Python's block statements (like if, for, while) do **not create a new scope**.



Summary of Scope Types

Scope Type	Where Defined	Accessible In
Local	Inside a function	Only that function
Global	Outside all functions	Everywhere in the file
Enclosing	Outer function of a nested function	Inner function
Built-in	Python provided	Anywhere



➤ Python Input – input() Function

□ Syntax:

```
variable = input("Prompt message")
```

◆ Always returns data as a string

□ Example:

```
name = input("Enter your name: ")  
print("Hello", name)
```

↻ Convert to int/float:

```
age = int(input("Enter age: "))  
price = float(input("Enter price: "))
```



➤ Output – print() Function

□ Syntax:

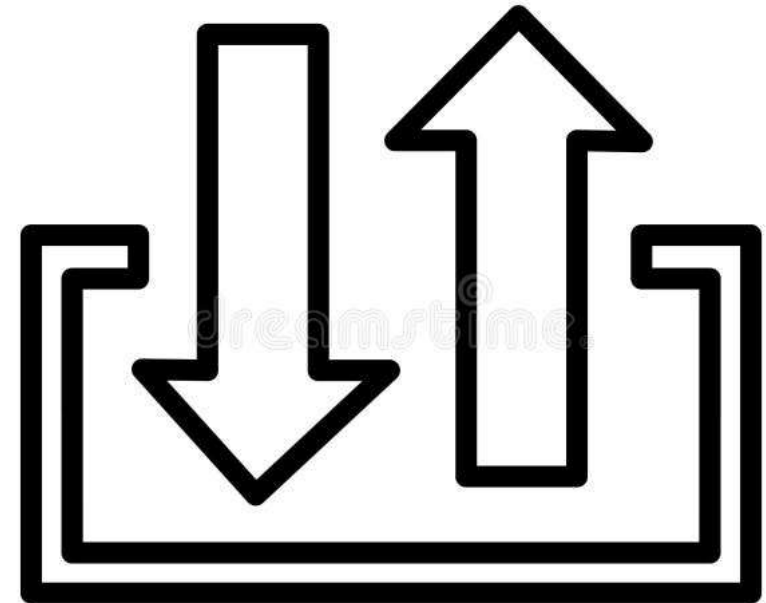
```
print(object, ..., sep=' ', end='\n')
```

□ Example:

```
print("Python", "is", "Fun", sep="-", end="!")
```

Output: Python-is-Fun!

✓ Multiple values can be printed at once



➤ Formatted Output – f-strings

- Introduced in Python 3.6

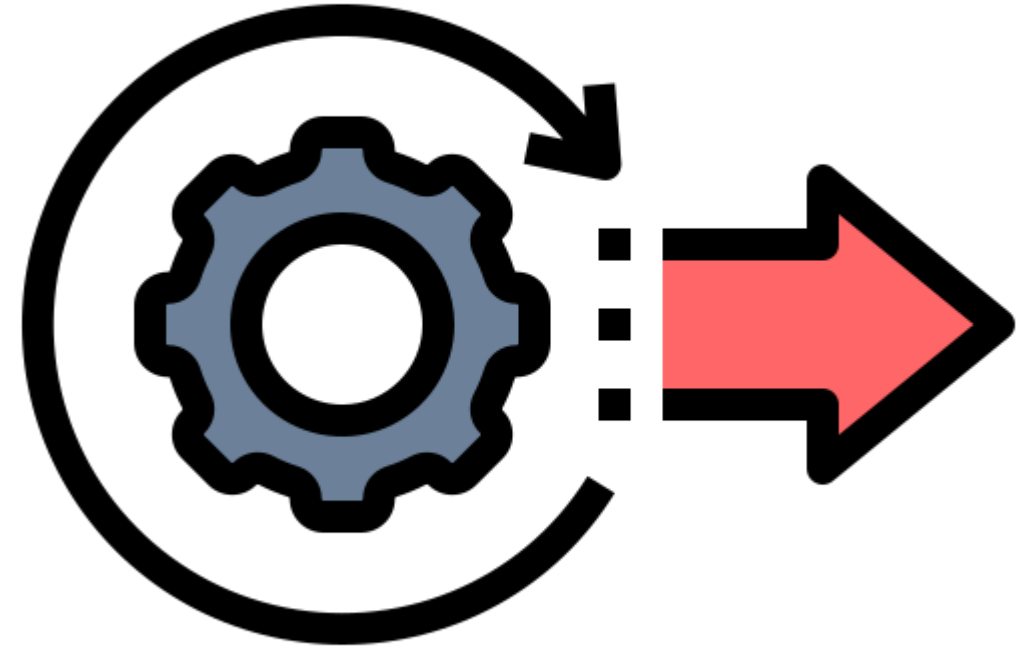
- ✦ Easier and more readable

- Example:

```
name = "Ritesh"  
age = 22  
print(f"My name is {name} and I am {age} years old.")
```

Alternate (Python 3+):

```
print("My name is {} and I am {} years old.".format(name, age))
```



➤ More Examples of Input/Output

📄 Example 1: Sum of two numbers

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
print("Sum is:", a + b)
```

📄 Example 2: Print Table

```
n = int(input("Enter number: "))  
for i in range(1, 11):  
    print(f"{n} x {i} = {n*i}")
```



Quick Recap / Takeaway

SCOPE:

- local: function level
- global: outside all functions
- enclosing: nested function
- block scope: not applicable in Python

INPUT:

- Use `input()`, always returns str

OUTPUT:

- Use `print()`
- Format using f-strings or `.format()`



Practice Questions / Exercises

1 Write a program that:

- Takes name and age as input
- Prints a greeting message

2 Create a nested function that prints an enclosing variable

3 Demonstrate modifying a global variable inside a function

4 Check if a block-scoped variable (inside if) is accessible outside



